Public repository copy – filtered, not the complete submission draft. Tissue images, individual-slide plots, case-level tables and row-level source data are withheld. Marked omissions preserve the original numbering and captions. References to Source Data describe the local research archive; only screened aggregate tables are distributed here. Reported aggregate estimates are unchanged.

Weakly supervised whole-slide classification of salivary gland MALT lymphoma versus Sjögren's syndrome

[MISSING A01: author names, affiliations and author order]

4 November 2026 | Draft v7.5

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Abstract

Background: Salivary gland MALT lymphoma and Sjögren's syndrome (SS) can share lymphoid-rich histological appearances. We evaluated a weakly supervised whole-slide classifier that combines multiple summaries of fixed pathology foundation-model features to distinguish these diagnostic groups.

Methods: The internal cohort contained 193 whole-slide images: 144 for training and 49 for validation. Hybrid-Evidence multiple-instance learning used fixed UNI2-h embeddings and a checkpoint selected by internal validation AUROC. External evaluation used the same checkpoint on 168 slides (152 MALT, 16 SS) with mixed archive stain tags awaiting confirmation. Comparisons included alternative MIL implementations, an exploratory score ensemble and simple feature-based controls.

Results: Hybrid achieved internal validation AUROC 0.9599 (95% CI 0.9046–0.9967), sensitivity 95.65% and specificity 80.77% at threshold 0.5. Its AUROC difference from ABMIL was 0.0100 (95% CI −0.0184 to 0.0452). The three-model ensemble achieved internal AUROC 0.9649. External Hybrid AUROC was 0.8721 (95% CI 0.7364–0.9758), with sensitivity 93.42% and specificity 68.75%. After post hoc exclusion of IHC/special-stain-tagged slides, AUROC was 0.8689 and sensitivity 75.61%; this was not verified H&E-only validation.

Conclusions: Hybrid-Evidence MIL supported retrospective MALT–SS discrimination using fixed pathology representations. Validation reuse, unverified patient-level separation and mixed external material limit clinical interpretation. Independent evaluation with verified cohort metadata is required before diagnostic utility can be assessed.

Keywords: salivary gland; MALT lymphoma; Sjögren's syndrome; whole-slide imaging; multiple-instance learning

1 Introduction

Salivary gland mucosa-associated lymphoid tissue (MALT) lymphoma and Sjögren's syndrome (SS)-associated inflammation can share lymphoid-rich histological appearances. SS is also a clinical setting for lymphomagenesis [1]. Lymphoepithelial lesions and architectural changes occur in both processes, making the composition and distribution of infiltrates relevant to pathological assessment [2]. The diagnostic distinction rests on integrated morphology and, when indicated, immunophenotypic and molecular findings within current lymphoma classification frameworks [3, 4]. An image-based classifier could provide an additional signal for review, but its diagnostic scope must be established in the relevant case spectrum.

Computational studies have addressed SS diagnosis using cell–tissue graphs and automated evaluation of minor salivary gland focus scores [5, 6]. Related lymphoma studies have distinguished orbital

MALT lymphoma from IgG4-related ophthalmic disease or classified MALT and diffuse large B-cell lymphoma in selected histological material [7, 8]. These studies differ in tissue, reference categories and image-sampling strategy. They motivate direct evaluation of salivary gland MALT-versus-SS classification without making their reported accuracy interchangeable with this task. Here, MALT and SS denote the recorded slide-level diagnostic groups; an SS medical history in a patient with MALT is a separate clinical attribute.

Weakly supervised multiple-instance learning (MIL) can learn slide-level classifications without exhaustive lesion annotations [9–11]. Pathology foundation models provide reusable patch representations, allowing disease-specific aggregation to be trained from cached embeddings [12, 13]. Recent benchmarking and the nnMIL framework emphasize that feature representations and training recipes must be considered alongside aggregation architecture [14, 15]. Simple feature-based controls and comparisons of complete implementations therefore provide useful context for evaluating a complex model on a small disease-specific cohort.

We studied retrospective MALT–SS classification using fixed UNI2-h embeddings and a custom Hybrid-Evidence MIL aggregator. The model combines attention, distributional and prototype-margin summaries to produce a slide-level score. We evaluated its discrimination against alternative MIL implementations, assessed the selected checkpoint on an external collection and examined model-selected regions. Because internal validation slides also selected checkpoints and the external material has mixed archive stain tags, we distinguish development estimates from independent clinical validation. Additional feature-based controls and implementation analyses accompany the primary results.

2 Results

2.1 Study material and validation design

The internal cohort comprised 193 whole-slide images (WSIs): 90 MALT and 103 SS. A label-stratified slide-level split assigned 144 WSIs to training (67 MALT, 77 SS) and 49 to internal validation (23 MALT, 26 SS; Table 1). The 49 validation slides were evaluated during training to select the checkpoint and were also used for the reported internal estimates. These estimates therefore describe model development, rather than performance on an independent internal test set. [MISSING C02: hospital-held patient linkage, patient counts, dates and clinical characteristics; patient-level separation remains unverified.]

The external archive contained 227 slides; 223 produced predictions and 168 entered the final author-reviewed analysis (152 MALT, 16 SS). Among the 152 MALT slides, 33 had H&E-like archive tags, 111 had IHC/special-stain tags and eight came from a separate HE directory. Stains in the 16 SS slides were not confirmed in the available metadata. These groups describe archive provenance, not verified stain-specific cohorts. [MISSING C03/C04: hospital-confirmed stain/specimen metadata and finalized eligibility and review criteria.]

Hybrid-Evidence MIL combined five summaries of 512 sampled UNI2-h patch embeddings with eight latent similarity/margin statistics (Fig. 1). The primary checkpoint was selected by the highest internal validation AUROC and used unchanged for external evaluation.

2.2 Discrimination on the internal validation set

Hybrid achieved an area under the receiver operating characteristic curve (AUROC) of 0.9599 (95% CI 0.9046–0.9967) and average precision (AUPRC) of 0.9514 (Fig. 2; Table 2). At threshold 0.5, it

Table 1: WSI cohort composition

Analysis set	Diagnostic/source group	WSIs	Verified patients
Internal training	MALT	67	[MISSING C02]
Internal training	SS	77	[MISSING C02]
Internal validation	MALT	23	[MISSING C02]
Internal validation	SS	26	[MISSING C02]
External selected	MALT: source B	144	[MISSING C02]
External selected	MALT: HE directory	8	[MISSING C02]
External selected	SS	16	[MISSING C02]

Both internal and external analyses compare MALT lymphoma with SS. [MISSING C02: add age, sex, clinical features, collection dates and verified patient counts. Hospital-held records; pending incorporation.]

correctly classified 22/23 MALT and 21/26 SS slides, giving sensitivity 95.65%, specificity 80.77% and accuracy 87.76%. The intervals condition on the saved predictions and do not correct for checkpoint selection on these slides.

Table 2: Internal validation estimates; threshold metrics at 0.5

Model	AUROC	AUPRC	Accuracy	Sensitivity	Specificity	F1
Three-model ensemble	**0.9649**	**0.9576**	87.76%	91.30%	84.62%	0.8750
Hybrid-Evidence MIL	0.9599	0.9514	87.76%	**95.65%**	80.77%	0.8800
ABMIL	0.9498	0.9396	85.71%	82.61%	88.46%	0.8444
DTFD-style	0.9398	0.9416	87.76%	86.96%	88.46%	0.8696
CLAM-SB-style	0.9365	0.9422	87.76%	91.30%	84.62%	0.8750
RRTMIL-style	0.9281	0.8400	**89.80%**	**95.65%**	84.62%	**0.8980**
ProtoMIL-style	0.9264	0.9306	85.71%	86.96%	84.62%	0.8511
Mean pooling	0.9247	0.9332	85.71%	78.26%	92.31%	0.8372
TransMIL-style	0.9231	0.9180	81.63%	65.22%	**96.15%**	0.7692
DSMIL-style	0.9197	0.9260	85.71%	78.26%	92.31%	0.8372
Distribution-ProtoMIL	0.9130	0.9178	79.59%	91.30%	69.23%	0.8077
CONCH v1.5 + TITAN	0.9013	0.9203	81.63%	91.30%	73.08%	0.8235

Bold and underlined values indicate the best and second-best distinct values in each metric column; ties share the same mark. Rankings are descriptive and do not imply statistical significance. n = 49 WSIs (23 MALT and 26 SS). Threshold-dependent metrics use 0.5. The same 49 slides selected checkpoints; these are development estimates. AUROC and AUPRC summarize the scores across thresholds. All models except CONCH v1.5 + TITAN use UNI2-h features. Slide-encoder dimensions, classifier inputs and optimization budgets differ; these are complete-implementation comparisons rather than isolated pooling substitutions. Interval estimates and predictive values appear in Supplementary Table S2. The ensemble at 0.70 is a separate archived operating point, not an additional model.

ABMIL achieved AUROC 0.9498, compared with 0.9247 for neural mean pooling. The paired Hybrid-minus-ABMIL AUROC difference was 0.0100 (95% CI −0.0184 to 0.0452; Supplementary Table S4). Hybrid alone correctly classified four discordant slides and ABMIL alone correctly classified three (exact McNemar $P = 1.000$). Other UNI2-h configurations achieved AUROCs of

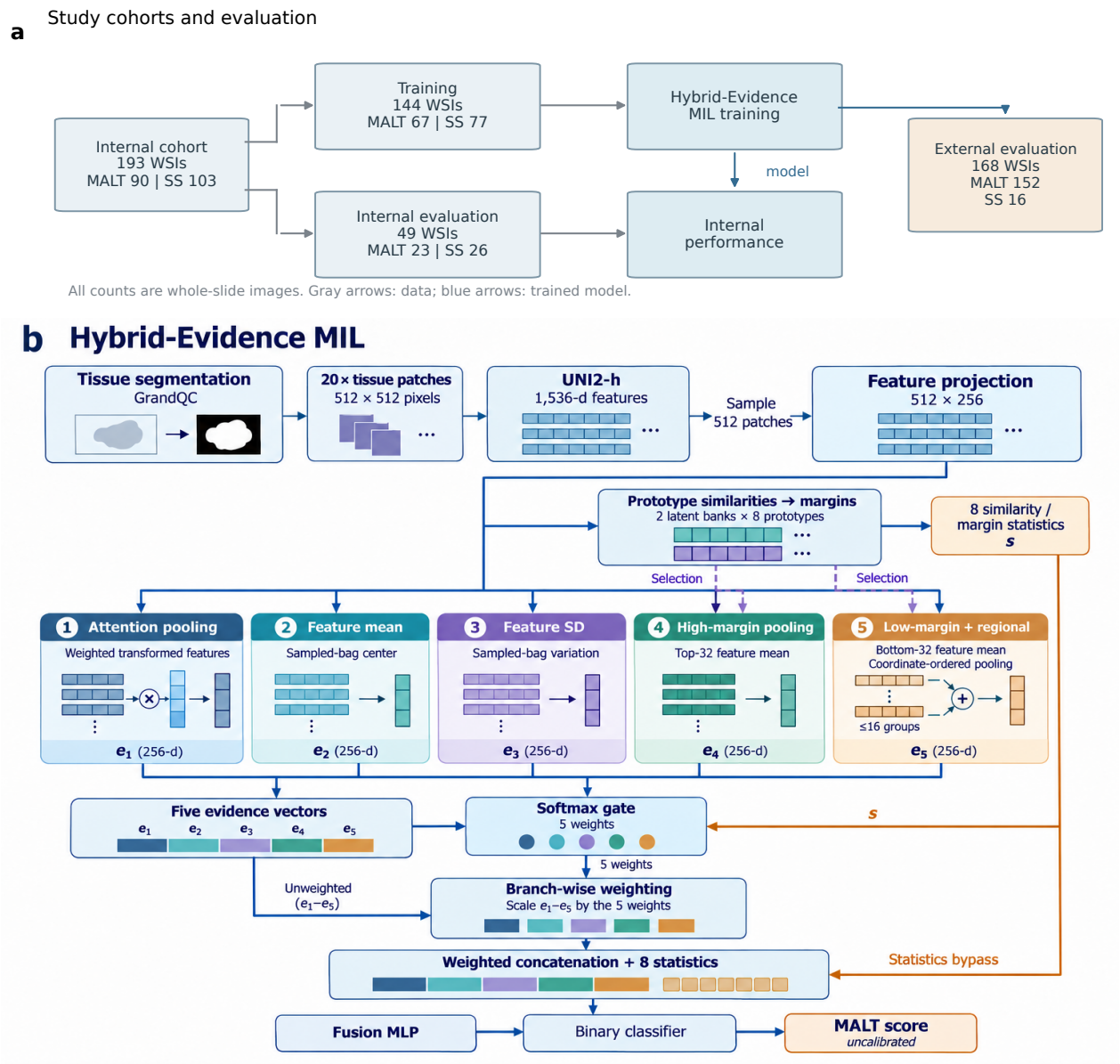


Figure 1: Study cohorts and multi-evidence classification. **a,** Internal training and validation cohorts and the external evaluation material; all counts are WSIs. The internal validation slides also selected checkpoints; patient-level separation remains unverified. Gray arrows indicate data and blue arrows indicate the trained model. **b,** Projected UNI2-h features form five evidence vectors. Similarities to two latent prototype banks and their margins generate instance selections and eight separate statistics. The low-margin instance and regional vectors are added; regional tokens are formed by coordinate-ordered grouping and pooled without explicit interregional spatial relations. Gate weights scale the five vectors before concatenation; the statistics also enter fusion through an unweighted pathway. Source data are provided in internal_cohort.csv and external_final_predictions.csv.

0.9130–0.9398. Their dimensions, classifier heads and optimization settings differed, so these are comparisons of complete local implementations (Supplementary Tables S1 and S14). CONCH v1.5 with TITAN achieved AUROC 0.9013 with a different encoder and patch budget.

The equal-weight Hybrid/CLAM-SB-style/RRTMIL-style score ensemble achieved AUROC 0.9649 and AUPRC 0.9576; its paired AUROC difference from Hybrid included zero. At threshold 0.5, accuracy remained 87.76%: the ensemble corrected one SS error but introduced one MALT error, with five errors shared. The retrospectively selected threshold of 0.70 gave accuracy 93.88%, sensitivity 91.30% and specificity 96.15% (Supplementary Table S3). Ensemble and threshold comparisons used

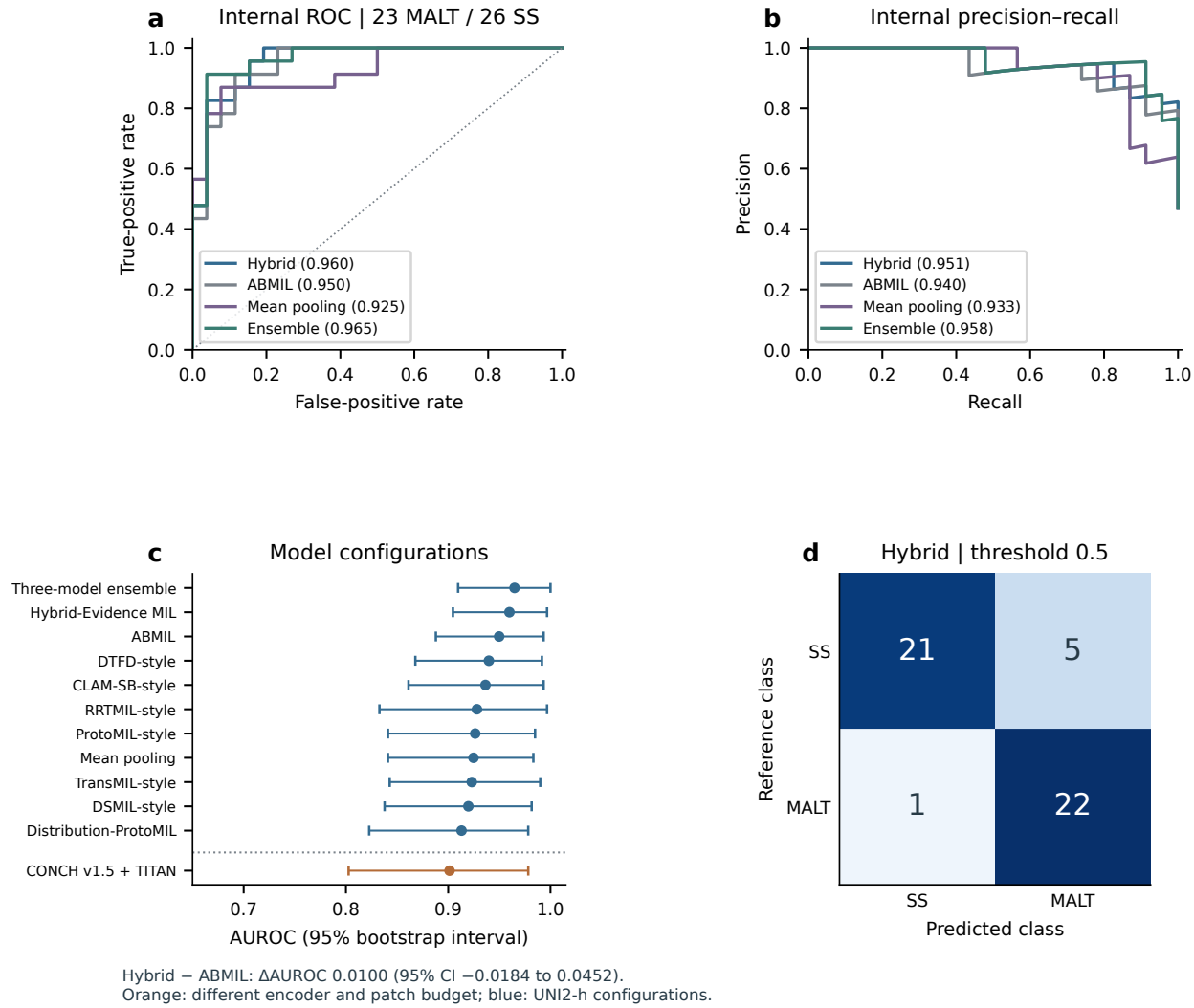


Figure 2: Internal validation discrimination and model configurations. **a,b,** ROC and precision–recall curves on 49 WSIs (23 MALT and 26 SS). **c,** AUROC and 95% intervals from 2,000 label-stratified WSI bootstrap samples of fixed predictions. Blue points use UNI2-h; the orange point uses a different encoder and patch budget. Simplified implementations retain their "-style" labels. The comparisons include complete implementations with different projection dimensions and training budgets. **d,** Hybrid confusion matrix at threshold 0.5. The 49 slides also selected checkpoints; intervals condition on fixed predictions and omit selection uncertainty. The paired Hybrid–ABMIL difference interval crosses zero. Source data are provided in internal_model_metrics.csv, paired_model_comparisons.csv and model-specific prediction tables.

the same internal validation material and are exploratory.

2.3 External discrimination and material composition

External Hybrid AUROC was 0.8721 (95% CI 0.7364–0.9758) and AUPRC was 0.9733 (Fig. 3; Table 3). At threshold 0.5, 142/152 MALT and 11/16 SS slides were correctly classified. Sensitivity was 93.42%, specificity 68.75% (95% CI 44.40%–85.84%) and balanced accuracy 81.09%. Accuracy was 91.07%, compared with 90.48% for a constant MALT prediction at the observed class prevalence. Ten of the 21 negative predictions were MALT, giving a negative predictive value of 52.38% (95% CI 32.37%–71.66%).

The archive-tag groups differed in their MALT detection counts: 23/33 H&E-like slides, 111/111

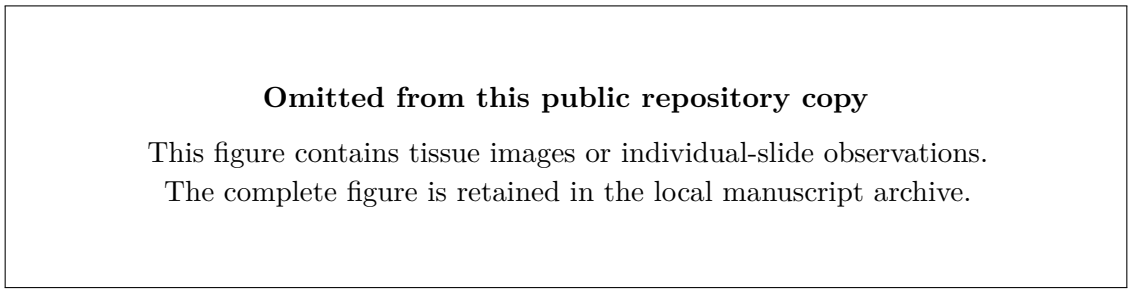


Figure 3: External performance and archive-tag composition. **a,b,** ROC and precision–recall curves for four frozen model checkpoints on the same 168 WSIs (152 MALT and 16 SS), using identical reference bags for each slide. Legend values indicate AUROC and average precision, respectively. These compare complete implementations. The dotted PR reference is the MALT fraction. **c,** Confusion matrix at threshold 0.5. **d,** Individual uncalibrated model scores by diagnostic group and archive stain tag. Counts indicate correct predictions divided by group size; black marks show medians. Jitter separates points only. H&E-like, IHC/special and HE-archive labels are provisional archive descriptors, not pathology-confirmed stains. Single-class groups have no within-group AUROC. Source data are provided in external_final_predictions.csv, external_archive_tag_counts.csv and reference_predictions.csv.

Table 3: Internal and external Hybrid estimates with 95% intervals

Cohort	Metric	Estimate (95% interval)
Internal Hybrid	AUROC	0.9599 (0.9046–0.9967)
Internal Hybrid	AUPRC	0.9514 (0.8864–0.9965)
Internal Hybrid	Accuracy	87.76% (75.76%–94.27%)
Internal Hybrid	Sensitivity	95.65% (79.01%–99.23%)
Internal Hybrid	Specificity	80.77% (62.12%–91.49%)
Internal Hybrid	PPV	81.48% (63.30%–91.82%)
Internal Hybrid	NPV	95.45% (78.20%–99.19%)
Internal Hybrid	F1	0.8800 (0.7925–0.9583)
External Hybrid	AUROC	0.8721 (0.7364–0.9758)
External Hybrid	AUPRC	0.9733 (0.9365–0.9971)
External Hybrid	Accuracy	91.07% (85.79%–94.51%)
External Hybrid	Sensitivity	93.42% (88.31%–96.39%)
External Hybrid	Specificity	68.75% (44.40%–85.84%)
External Hybrid	PPV	96.60% (92.29%–98.54%)
External Hybrid	NPV	52.38% (32.37%–71.66%)
External Hybrid	F1	0.9498 (0.9257–0.9732)

AUROC, average precision and F1 intervals use 2,000 label-stratified WSI bootstrap samples. Proportion intervals use Wilson intervals. Threshold-dependent metrics use 0.5. Internal slides also selected checkpoints; fixed-score intervals omit model-selection uncertainty and unverified patient clustering.

IHC/special-stain-tagged slides and 8/8 HE-directory slides were positive. All ten MALT false negatives occurred in the H&E-like group. A post hoc sensitivity analysis excluding the 111 IHC/special-stain-tagged slides retained 57 WSIs (41 MALT, 16 SS). AUROC was 0.8689, but sensitivity was 75.61% and balanced accuracy 72.18%; specificity remained 68.75% (Supplementary Table S18). This calculation shows that the high overall sensitivity depended on the evaluated material composition. It does not isolate a stain effect or establish H&E-only validation, because stain identities and the retained SS material remain unverified.

On identical external reference bags, the historical ABMIL, CLAM-SB-style and RRTMIL-style

checkpoints achieved AUROCs of 0.7903, 0.6591 and 0.7743, respectively (Supplementary Table S11). Paired Hybrid-minus-comparator differences were 0.0818 (95% CI −0.0136 to 0.2011), 0.2130 (0.0843–0.3528) and 0.0979 (0.0415–0.1649). These comparisons concern retained complete implementations; the historical three-model ensemble was not evaluated externally.

Hybrid Brier scores were 0.0895 internally and 0.0919 externally (Supplementary Table S13). For context, the external-prevalence constant had squared error 0.0862 and the training-prevalence constant had squared error 0.2793 (Supplementary Table S18). The former uses evaluation labels and is only a descriptive reference. Neither the Brier score nor the high external AUPRC establishes calibration or clinical probability accuracy.

2.4 Additional model comparisons

Train-only logistic regression on mean UNI2-h features achieved internal/external AUROCs of 0.9314/0.8335; averaging normalized patch features gave 0.9247/0.8462. Their external sensitivities were 45.39% and 46.05%, respectively, at specificity 93.75% for both (Supplementary Table S17). These controls retained discrimination with a different threshold tradeoff. Additional configuration comparisons and complete development results are reported in Supplementary Tables S9 and S15–S19 and Supplementary Results. They did not establish an advantage attributable to the learned gate.

2.5 Provisional reader observations

Aggregate records for two pathologists on the 49 internal slides gave accuracies of 77.55% and 83.67% (Fig. 4; Table 4). Reader 1 had 17 true positives, six false negatives, 21 true negatives and five false positives; Reader 2 had 19, four, 22 and four, respectively. These counts remain provisional pending reconciliation with individual decisions and the review protocol. [MISSING R01/R02: hospital-held reader records, experience and review conditions; paired analyses pending.]

The model estimates used a validation set that had participated in model selection. Together with the missing paired reader records, this precludes a reader-superiority claim from the displayed comparison. No AI-assisted reading benefit was measured by these aggregate counts.

Table 4: Draft reader comparison with evidence status

Reader/model	Evidence status	Threshold	Accuracy	Sensitivity	Specificity
Reader 1	Provisional aggregate	Not applicable	77.55%	73.91%	80.77%
Reader 2	Provisional aggregate	Not applicable	83.67%	82.61%	**84.62%**
Hybrid	Model predictions	0.5	**87.76%**	**95.65%**	80.77%
Three-model ensemble	Model predictions	0.5	**87.76%**	91.30%	**84.62%**

[MISSING R01/R02: verify reader aggregates using individual records and complete paired comparisons.]
Bold/underline mark the highest/second-highest displayed value in each performance column, with ties marked equally. Reader rankings are provisional descriptive comparisons. Both model rows use threshold 0.5. The ensemble operating point at 0.70 appears separately in Figure 4b and Supplementary Table S3. The internal model estimates reuse slides involved in checkpoint selection. These results do not establish reader superiority or AI-assisted improvement.

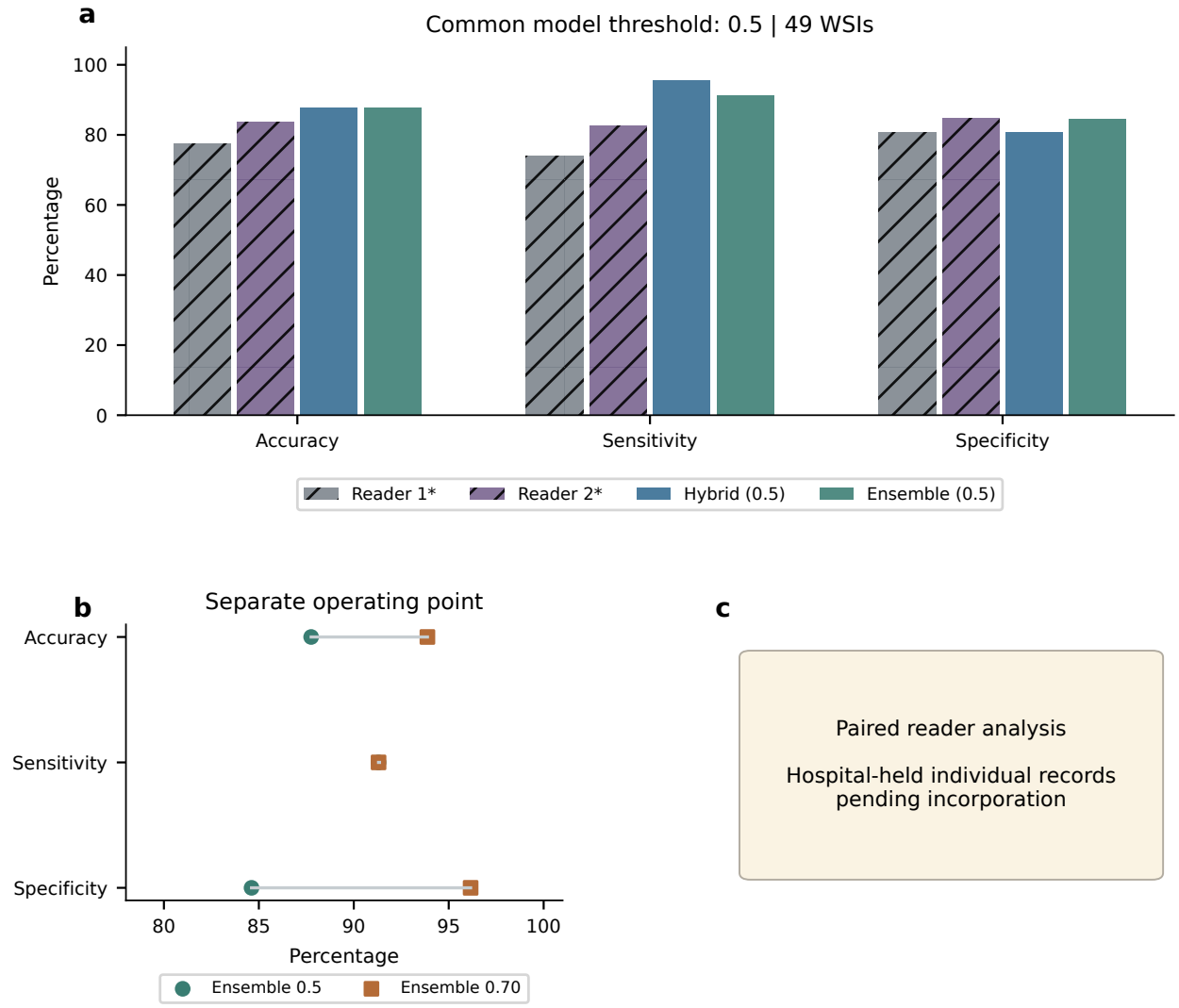


Figure 4: Reader comparison and separate model operating points. **a,** Provisional reader aggregates (hatched) and both models at threshold 0.5 on the 49-slide internal set. **b,** The ensemble at 0.5 and the retrospectively selected 0.70 operating point. **c,** Paired reader analyses remain unfilled pending incorporation of hospital-held records. Reader values are not assigned error bars or significance claims before reconciliation. Model estimates reuse validation slides involved in selection; no superiority inference is made. Source data are provided in reader_aggregate_PROVISIONAL.csv, internal_model_metrics.csv and ensemble_threshold_0_70.json. [MISSING R01/R02: hospital-held individual reader records and subsequent paired analysis.]

2.6 Inspectable regions and input-sampling variability

Figure 5 shows correct and incorrect classifications with their original model-selected windows. Main-evaluation and interpretation runs sampled different bags; both scores are linked in Supplementary Table S6. None of the 12 displayed interpretation cases changed class at threshold 0.5, although the largest score difference was 0.3252. All eight displayed thumbnails matched their original image windows. Automated segmentation-contour coverage was 100% for seven windows and 48.11% for the low-margin window in Case 10 (Supplementary Table S12); this geometric measure is not verified tissue content.

The overlays depict latent prototype margins normalized separately by slide. The mean-feature

branch had the largest gate coefficient on all 217 reference bags. Neither the latent banks nor gate coefficients establish disease-specific morphology or causal feature importance. The representative photomicrographs in Supplementary Fig. S2 provide pathological context, rather than annotations of these selected regions. [MISSING P01/P03: specimen linkage and calibration of illustrative photomicrographs, and pathological assessment of model-selected regions.]

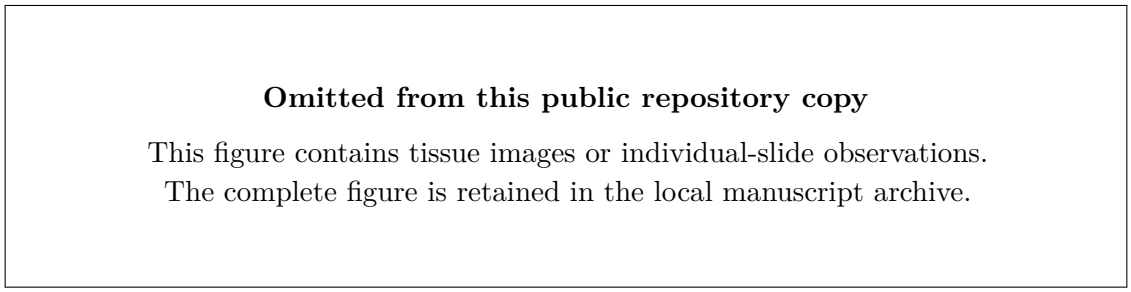


Figure 5: Model-selected regions and internal coefficients in correct and incorrect classifications. **a–d,** Cases 1, 4, 7 and 10 represent a true positive, true negative, false positive and false negative, respectively. Both main-evaluation and interpretation-run uncalibrated scores are shown. Overlays encode prototype margin, not attention; colors show margin divided by a per-slide scale, with saturation beyond ±1. Colors encode the difference between latent-bank similarities and have no independently established disease-specific morphological meaning. Color intensity is not directly comparable between cases. H/L locators identify the highest/lowest-margin thumbnails, verified by exact pixel matching to the original sampled windows. Scale bars, 50 μm; source windows span approximately 231.1 μm. **e,** Gate coefficients scale feature vectors and do not establish causal importance. All eight original thumbnails, including the low-margin crop in Case 10, are retained. Archived segmentation-contour coverage is summarized in Supplementary Table S12; it does not establish pathology-confirmed tissue or lesion content. Case 10 locators preserve the historical overlay extent documented in that table. Source data include interpretation_probability_comparison.csv and interpretation_gate_weights.csv. [MISSING P03: pathology-confirmed regional descriptions.]

Across 20 additional bags per slide with the historical checkpoint fixed, median score standard deviations were 0.0120 internally and 0.0150 externally; maxima were 0.1297 and 0.1241 (Fig. 6; Supplementary Table S10). Seven of 49 internal slides and 18 of 168 external slides received both threshold classifications across draws. Complete-cohort AUROC ranged from 0.9498 to 0.9682 internally and from 0.8549 to 0.8943 externally. These are technical sampling ranges. Averaging 20 scores per slide gave AUROCs of 0.9632 and 0.8775; externally, it traded one fewer false positive for one additional false negative. The original single-bag predictions remain the primary results.

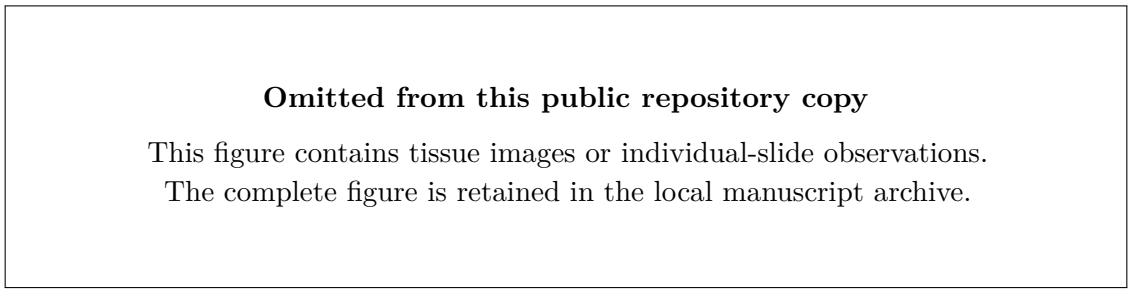


Figure 6: Fixed-checkpoint sampling variability and ensemble error exchange. **a,** Sample standard deviation of the uncalibrated Hybrid score across 20 additional 512-patch bags for each of 49 internal and 168 external WSIs; each dot is one WSI and horizontal marks denote medians. **b,** AUROC for each complete-cohort draw (20 dots per cohort). Horizontal marks denote draw medians; open diamonds denote the historical reference-bag AUROC. The spread across draws describes sampling variability, not a confidence interval. **c,** WSIs with at least one score below 0.5 and at least one score at or above 0.5 across the 20 bags, stratified by cohort and class. Labels give numerator/denominator. **d,** Correctness of archived Hybrid and ensemble predictions at threshold 0.5 on the same 49 internal WSIs. The ensemble corrects one SS error and introduces one MALT error, with five shared errors. Sampling repeats do not constitute new WSIs or retraining replicates. Source data: repeated_bag_predictions.csv and the archived model-specific internal predictions.

3 Discussion

Hybrid-Evidence MIL distinguished MALT lymphoma from SS in the evaluated salivary gland material using slide-label training and fixed UNI2-h features. The internally selected checkpoint achieved internal validation AUROC 0.9599 and external AUROC 0.8721. Combining patch-level summaries provided a practical way to evaluate this diagnostic distinction without exhaustive regional annotations. These results motivate further assessment of the classification signal in cohorts with verified patient linkage and specimen metadata.

Hybrid had the highest internal AUROC among the evaluated individual models, while score averaging produced the highest ensemble AUROC. This descriptive ranking does not establish superiority over ABMIL: the paired AUROC-difference intervals included zero in both cohorts. Mean-feature logistic regression also retained external discrimination, with lower sensitivity at threshold 0.5. Representation quality, optimization and aggregation therefore need to be considered together, as emphasized in recent pathology benchmarks and nnMIL [14, 15]. The complete implementation comparisons support the reported performance profile, while the supplementary configuration analyses do not establish an isolated benefit of the learned gate.

External composition materially affected the clinical interpretation. Only 16 SS slides informed specificity, and MALT represented 90.48% of the evaluated set. The reported accuracy of 91.07% was therefore close to a constant-positive classifier. Excluding IHC/special-stain archive tags left AUROC similar but reduced sensitivity from 93.42% to 75.61%. This does not prove that staining caused the difference: source, diagnosis, stain and specimen characteristics were not separated by the available design. It does show why a single mixed-material sensitivity cannot stand for verified H&E performance. The low external negative predictive value also prevents a rule-out interpretation. Class-weighted training and the absence of calibration fitting further limit interpretation of the sigmoid score as an individual probability.

The principal design limitation is the absence of an untouched internal test set. The same 49 slides selected epochs and supported model, ensemble and threshold comparisons. Fixed-score bootstrap intervals quantify uncertainty conditional on those predictions, not uncertainty from model selection. The split was performed by slide, and hospital-held patient linkage is needed to verify that patients

did not span partitions or contribute correlated observations. Later analyses also reused an external collection whose historical outcomes were already known. Locking candidate selection before new external inference constrains that development exercise, but does not turn the collection into a fresh confirmatory cohort.

The region displays make selected image content traceable, while leaving biological interpretation open. Prototype margins are internal similarity differences, and coordinate-group pooling does not encode explicit interregional spatial relationships. The largest gate coefficient need not identify the feature with the largest causal contribution. Regional pathological annotations would be required to connect these outputs with lymphoepithelial lesions, glandular architecture or artifacts. Work such as SMMILe evaluates regional predictions against spatial evidence [16]; a heatmap alone supplies a different level of evidence. Similarly, reader aggregates cannot establish an assisted-review benefit, which requires a defined reading protocol and individual paired outcomes [17, 18].

The next clinical evaluation should first reconcile patient identifiers, diagnoses, source/stain metadata and eligibility decisions, then use patient-disjoint development with a frozen model and operating point assessed in an untouched, appropriately specified cohort. More SS controls are needed for precise specificity estimates. A protocolized reader study and annotation of selected regions would address clinical contribution and morphological relevance. These steps would test the application suggested by the present discrimination results without assuming that model development has already established diagnostic utility.

4 Conclusion

Hybrid-Evidence MIL combined fixed pathology foundation-model features to distinguish salivary gland MALT lymphoma from SS, with internal validation AUROC 0.9599 and external AUROC 0.8721. These retrospective results support further evaluation of the approach. Patient-level verification and independent assessment of a frozen model in a defined specimen cohort are required before diagnostic utility can be established.

5 Methods

5.1 Study design, reference categories and ethics

This retrospective study used WSI-level diagnostic labels, with MALT positive and SS negative. Source records identified Peking University School and Hospital of Stomatology as the internal source and Peking University Third Hospital as an external source. [AUTHOR CHECK C05: confirm the provenance of the separate HE-directory material.] Hospital-held records are needed to characterize the actual diagnostic spectrum, SS histories, reference-standard procedures, reviewers and ancillary tests. The relevant lymphoma and SS classification frameworks provide context but are not substituted for the procedures actually used [3, 4, 19]. [MISSING C02/C06: patient linkage, dates, clinical characteristics and reference-standard details.]

The Biomedical Ethics Committee of Peking University School and Hospital of Stomatology approved the protocol (PKUSSIRB-2025110092; review on 24 April 2025, signature on 27 May 2025). [AUTHOR CHECK A03: insert consent or waiver and external-material/continuing-review provisions from the hospital records.] Reporting was organized around STARD-AI [20].

5.2 Cohorts and development partitions

Slide identifiers were generated from file stems. Identifiers occurring in both diagnostic classes were removed; repeated same-label identifiers were deduplicated. Slides without the required precomputed features were excluded from the corresponding analysis. The main UNI2-h dataset contained 193 WSIs. A label-stratified split with holdout fraction 0.25 produced 144 training and 49 validation slides. The split code did not group slides by patient; patient-level separation cannot be verified without linkage records.

For the external material, original archive membership (227 slides), prediction success (223) and final author-reviewed inclusion (168) were recorded separately. [MISSING C04: finalized eligibility, exclusion and manual-review criteria.] Source-B MALT tags used filename keywords for H&E-like material and assigned remaining slides to IHC/special stain; the separate HE directory and SS series retained their source descriptors. [MISSING C03: pathology-confirmed stains and specimen metadata.] No archive descriptor was treated as a verified stain identity.

5.3 Image processing and frozen representations

TRIDENT preprocessing used GrandQC tissue segmentation [21]. All 361 analyzed WSI headers recorded a Hamamatsu C9600-12 scanner, 40× source magnification and approximately 0.225708 μm per level-0 pixel. Non-overlapping 512×512 patches at target 20× corresponded to $1{,}024 \times 1{,}024$ level-0 windows. UNI2-h produced 1,536-dimensional features stored with patch coordinates; the image encoder remained frozen during downstream training [12, 22]. Its recorded image transform resized and centre-cropped to 224×224 pixels. Detailed encoder settings and recovered acquisition metadata appear in Supplementary Methods and Table S7. [MISSING M01: exact downloaded UNI2-h revision and weight checksum.]

Each neural input bag contained 512 feature rows, sampled without replacement when possible and with replacement otherwise. Training bags varied across loading events; evaluation used recorded deterministic sampling conventions. CONCH v1.5 with TITAN used 768-dimensional features and 1,024 patches, and was treated as a separate encoder/aggregation comparison [23].

5.4 Hybrid model and historical comparisons

Hybrid projected patch features from 1,536 to 256 dimensions with layer normalization, GELU and dropout 0.3. Five vectors summarized gated attention, the feature mean, sample standard deviation, high-margin instances, and the sum of low-margin instances and pooled coordinate groups. Two banks of eight learnable prototypes defined relative cosine-similarity margins; the 32 highest- and lowest-margin instances formed their corresponding summaries. Six continuous similarity/margin statistics and two indicator proportions provided eight additional scalar inputs. Sixteen coordinate-ordered groups were pooled without explicit intergroup distances or adjacency.

A learned five-output gate scaled the vectors before concatenation. The eight statistics entered the fusion network through an unweighted path, followed by a classifier and sigmoid score. Prototype banks were latent parameters without regional diagnostic labels. Gate coefficients represented feature scaling, not causal attribution. Full equations and dimensions are given in Supplementary Methods.

The historical Hybrid used AdamW, learning rate 5×10^{-5}, weight decay 10^{-4}, batch size eight and 30 epochs. Weighted binary cross-entropy used positive-class weight 77/67. The highest internal validation AUROC selected epoch 15. Thus, the validation set was repeatedly inspected during training and subsequently used to report discrimination.

Historical comparators included neural mean pooling, ABMIL and local CLAM-SB-, DSMIL-, DTFD-, TransMIL-, RRTMIL- and prototype-based implementations [9, 11, 24–27]. The "-style" designation identifies compact local variants, not complete official reproductions; CLAM-SB-style omitted instance clustering, and the local RRTMIL-style implementation does not reproduce the full feature re-embedding method. Dimensions and training budgets differed (Supplementary Tables S1 and S14). The exploratory ensemble averaged the three scores from Hybrid, CLAM-SB-style and RRTMIL-style. Its threshold of 0.70 was selected retrospectively on the internal validation material; 0.5 is the common display threshold.

5.5 Additional model comparisons

Supplementary configuration analyses examined feature normalization, attention masking, direct/subspace attention and two-bag training under a recorded development protocol. Checkpoints and candidate selection used internal validation AUROC; the complete training settings, selection records and outcomes are provided in Supplementary Methods and Tables S15–S19. These analyses reused the existing cohorts.

Two simple controls fitted train-only standardization and L2-regularized logistic regression to all-feature slide means, with or without per-patch L2 normalization. They used $C = 1$, balanced class weights and no hyperparameter search. Reference-bag inference and full-feature means were evaluated separately. These controls are distinct from the historical neural mean-pooling model.

5.6 Secondary analyses, readers and regional displays

Historical external comparator evaluation used identical reconstructed reference bags for four frozen checkpoints. Twenty additional bags per slide quantified the historical Hybrid checkpoint's sampling variation; a threshold-crossing slide had predictions on both sides of 0.5. Draw-level AUROCs and score averages were descriptive technical analyses. The archive-tag sensitivity analysis removed the 111 IHC/special-stain-tagged MALT slides without refitting or changing the threshold; it was post hoc.

Reader material consisted of provisional aggregate counts. Individual decisions, reference labels, experience, randomization, blinding and presentation conditions await hospital record incorporation. Paired comparisons and assisted-reader outcomes were not inferred from aggregates. Region displays reused original images and linked main-evaluation and interpretation scores; their sampled bags differed. Exact crop matching and automated contour overlap checked image provenance, not pathological correctness. No local image enhancement or replacement of selected tissue was performed. Supplementary Methods document sampling and geometry.

5.7 Statistical analysis

The analysis unit was the WSI. AUROC measured rank discrimination; AUPRC denotes average precision [28]. Threshold metrics used 0.5 unless explicitly labeled otherwise. Balanced accuracy averaged sensitivity and specificity. Brier score was mean squared score error, with constant predictions based on training or evaluation prevalence supplied as descriptive references; the evaluation-prevalence reference uses evaluation labels.

Original AUROC, average-precision and F1 intervals used 2,000 class-stratified WSI bootstrap samples; proportion intervals used Wilson's method [29]. Additional paired external AUROC intervals used 10,000 class-stratified bootstrap samples, resampling the same slides across models.

Exact McNemar tests assessed discordant threshold decisions. Intervals and comparisons are descriptive, unadjusted for multiplicity, and conditional on fixed predictions; they do not incorporate model selection, patient clustering, prior external reuse, or training/bag variation. No unprovided patient grouping was inferred. Random-state records and supplementary training summaries are provided in Supplementary Methods.

Metric regeneration used Python, NumPy, pandas, SciPy and scikit-learn. Original figure assets were retained in this revision. Full runtime versions, hashes and supplementary implementation details accompany the source materials. [MISSING M02: original training hardware and historical dependency versions.]

Declarations

Data availability. Numeric Source Data accompany the manuscript as a separate archive. They comprise publication-coded slide-level model predictions, internal split labels, model and component summaries, interpretation-case outputs and acquisition metadata recovered from the WSI headers. Additional data include matched external comparator scores, repeated-bag outputs and exact indices, model-variable diagnostics, paired error summaries, ROI audit measurements, additional model-development results and two logistic controls. These include complete run outcomes, selection records and secondary aggregation analyses. Original slide filenames and filesystem paths are omitted from the coded prediction tables. The reader file contains provisional aggregate counts only. The contents and access status are summarized in Supplementary Table S8. [MISSING A04: confirm the institution-approved public release scope, repository identifier and access route for raw WSIs and clinical records.] The accompanying draft archive is not a public repository deposit.

Code availability. Archived custom training and external-inference source files, the main model configuration, the fixed-checkpoint, controlled-training and bounded-development protocols and runners, and metric/figure-generation scripts accompany the manuscript in a Source Code archive. File identities are documented by SHA-256 hashes. The archive records the implementation and retains the original script path configuration; it is not a newly validated portable training release. The new checkpoint analyses reproduced the available original reference scores and record their current runtime dependencies separately. Their input manifests require approved feature-file access; successful LaTeX compilation, statistical regeneration, checkpoint inference and complete training reproduction are distinct levels of reproducibility. TRIDENT is available from https://github.com/mahmoodlab/TRIDENT, and the UNI2-h model source is https://huggingface.co/MahmoodLab/UNI2-h. Foundation-model and trained-model weights are not redistributed in the draft archive. [MISSING A05: supply the author-controlled code repository, release licence and trained Hybrid weight-access arrangement.]

Ethics statement. The study protocol was approved by the Biomedical Ethics Committee of Peking University School and Hospital of Stomatology (approval No. PKUSSIRB-2025110092). The approval letter was signed on 27 May 2025, following review on 24 April 2025. [AUTHOR CHECK A03: insert the informed-consent or waiver statement and applicable external-material and continuing-review provisions from the hospital records.]

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Competing interests. [AUTHOR CHECK A08: confirm the final declaration for all authors.]

References

[1] Gaetane Nocturne, Elena Pontarini, Michele Bombardieri, and Xavier Mariette. Lymphomas complicating primary Sjögren's syndrome: from autoimmunity to lymphoma. *Rheumatology*, 60(8):3513–3521, 2021. doi: 10.1093/rheumatology/kez052. URL https://doi.org/10.1093/rheumatology/kez052.

[2] Yanting Chi, Qiulu Zhang, Zhiming Qin, Jiaying Bai, Jing Yan, Cuiling Liu, et al. Molecular pathology assists the diagnosis of lymphoepithelial sialadenitis, Sjögren's syndrome and extranodal marginal zone lymphoma of mucosa-associated lymphoid tissue. *Journal of Dental Sciences*, 19(1):130–138, 2024. doi: 10.1016/j.jds.2023.05.018. URL https://doi.org/10.1016/j.jds.2023.05.018.

[3] Rita Alaggio, Catalina Amador, Ioannis Anagnostopoulos, Ayoma D. Attygalle, Iguaracyra Barreto de Oliveira Araujo, Emilio Berti, et al. The 5th edition of the World Health Organization Classification of Haematolymphoid Tumours: Lymphoid Neoplasms. *Leukemia*, 36(7):1720–1748, 2022. doi: 10.1038/s41375-022-01620-2. URL https://doi.org/10.1038/s41375-022-01620-2.

[4] Elias Campo, Elaine S. Jaffe, James R. Cook, Leticia Quintanilla-Martinez, Steven H. Swerdlow, Kenneth C. Anderson, et al. The International Consensus Classification of Mature Lymphoid Neoplasms: a report from the Clinical Advisory Committee. *Blood*, 140(11):1229–1253, 2022. doi: 10.1182/blood.2022015851. URL https://doi.org/10.1182/blood.2022015851.

[5] Ruifan Wu, Zhipei Chen, Jiali Yu, Peng Lai, Xuanyi Chen, Anjia Han, et al. A graph-learning based model for automatic diagnosis of Sjögren's syndrome on digital pathological images: a multicentre cohort study. *Journal of Translational Medicine*, 22(1):748, 2024. doi: 10.1186/s12967-024-05550-8. URL https://doi.org/10.1186/s12967-024-05550-8.

[6] Konstantinos N. Panagiotopoulos, Nikos Tsiknakis, Dimitrios I. Zaridis, Clio P. Mavragani, Athanasios G. Tzioufas, Dimitrios I. Fotiadis, and Andreas V. Goules. Evaluation of minor labial salivary gland focus score in Sjögren's disease using deep learning: a tool for more efficient diagnosis and future tissue biomarker discovery. *Journal of Autoimmunity*, 153:103418, 2025. doi: 10.1016/j.jaut.2025.103418. URL https://doi.org/10.1016/j.jaut.2025.103418.

[7] Mizuki Tagami, Mizuho Nishio, Atsuko Yoshikawa, Norihiko Misawa, Atsushi Sakai, Yusuke Haruna, et al. Artificial intelligence-based differential diagnosis of orbital MALT lymphoma and IgG4 related ophthalmic disease using hematoxylin–eosin images. *Graefe's Archive for Clinical and Experimental Ophthalmology*, 262(10):3355–3366, 2024. doi: 10.1007/s00417-024-06501-1. URL https://doi.org/10.1007/s00417-024-06501-1.

[8] Shuto Yamaguchi, Teijiro Isokawa, Nobuyuki Matsui, Naotake Kamiura, and Tatsuaki Tsuruyama. AI system for diagnosing mucosa-associated lymphoid tissue lymphoma and diffuse large B cell lymphoma using ImageNet and hematoxylin and eosin–stained specimens. *PNAS Nexus*, 4(5):pgaf137, 2025. doi: 10.1093/pnasnexus/pgaf137. URL https://doi.org/10.1093/pnasnexus/pgaf137.

[9] Maximilian Ilse, Jakub Tomczak, and Max Welling. Attention-based Deep Multiple Instance Learning. In *Proceedings of Machine Learning Research*, volume 80, pages 2127–2136, 2018. doi: 10.48550/arXiv.1802.04712. URL https://doi.org/10.48550/arXiv.1802.04712.

[10] Gabriele Campanella, Matthew G. Hanna, Luke Geneslaw, Allen Miraflor, Vitor Werneck Krauss Silva, Klaus J. Busam, et al. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nature Medicine*, 25(8):1301–1309, 2019. doi: 10.1038/s41591-019-0508-1. URL https://doi.org/10.1038/s41591-019-0508-1.

[11] Ming Y. Lu, Drew F. K. Williamson, Tiffany Y. Chen, Richard J. Chen, Matteo Barbieri, and Faisal Mahmood. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature Biomedical Engineering*, 5(6):555–570, 2021. doi: 10.1038/s41551-020-00682-w. URL https://doi.org/10.1038/s41551-020-00682-w.

[12] Richard J. Chen, Tong Ding, Ming Y. Lu, Drew F. K. Williamson, Guillaume Jaume, Andrew H. Song, et al. Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30(3): 850–862, 2024. doi: 10.1038/s41591-024-02857-3. URL https://doi.org/10.1038/s41591-024-02857-3.

[13] Eugene Vorontsov, Alican Bozkurt, Adam Casson, George Shaikovski, Michal Zelechowski, Kristen Severson, et al. A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30(10):2924–2935, 2024. doi: 10.1038/s41591-024-03141-0. URL https://doi.org/10.1038/s41591-024-03141-0.

[14] Rohan Bareja, Francisco Carrillo-Perez, Yuanning Zheng, Marija Pizurica, Tarak Nath Nandi, Lu Tian, Jeanne Shen, Ravi Madduri, and Olivier Gevaert. A benchmark study of vision and pathology foundation models for computational pathology. *Nature Communications*, 17:9012, 2026. doi: 10.1038/s41467-026-76004-6. URL https://doi.org/10.1038/s41467-026-76004-6.

[15] Xiangde Luo, Jinxi Xiang, Yuanfeng Ji, and Ruijiang Li. nnMIL: a generalizable multiple instance learning framework for computational pathology. *Nature Biomedical Engineering*, 2026. doi: 10.1038/s41551-026-01767-8. URL https://doi.org/10.1038/s41551-026-01767-8. Published online 25 August 2026.

[16] Zeyu Gao, Anyu Mao, Yuxing Dong, Hannah Clayton, Jialun Wu, Jiashuai Liu, ChunBao Wang, Kai He, Tieliang Gong, Chen Li, and Mireia Crispin-Ortuzar. SMMILe enables accurate spatial quantification in digital pathology using multiple-instance learning. *Nature Cancer*, 6(12):2025–2041, 2025. doi: 10.1038/s43018-025-01060-8. URL https://doi.org/10.1038/s43018-025-01060-8.

[17] Wouter Bulten, Kimmo Kartasalo, Po-Hsuan Cameron Chen, Peter Ström, Hans Pinckaers, Kunal Nagpal, et al. Artificial intelligence for diagnosis and Gleason grading of prostate cancer: the PANDA challenge. *Nature Medicine*, 28(1):154–163, 2022. doi: 10.1038/s41591-021-01620-2. URL https://doi.org/10.1038/s41591-021-01620-2.

[18] Wouter Bulten, Maschenka Balkenhol, Jean-Joël Awoumou Belinga, Américo Brilhante, Aslı Çakır, Lars Egevad, et al. Artificial intelligence assistance significantly improves Gleason grading of prostate biopsies by pathologists. *Modern Pathology*, 34(3):660–671, 2021. doi: 10.1038/s41379-020-0640-y. URL https://doi.org/10.1038/s41379-020-0640-y.

[19] Caroline H. Shiboski, Stephen C. Shiboski, Raphaèle Seror, Lindsey A. Criswell, Marc Labetoulle, Thomas M. Lietman, et al. 2016 American College of Rheumatology/European League Against Rheumatism Classification Criteria for Primary Sjögren's Syndrome: A Consensus and Data-Driven Methodology Involving Three International Patient Cohorts. *Arthritis & Rheumatology*, 69(1):35–45, 2017. doi: 10.1002/art.39859. URL https://doi.org/10.1002/art.39859.

[20] Viknesh Sounderajah, Ahmad Guni, Xiaoxuan Liu, Gary S. Collins, Alan Karthikesalingam, Sheraz R. Markar, et al. The STARD-AI reporting guideline for diagnostic accuracy studies using artificial intelligence. *Nature Medicine*, 31(10):3283–3289, 2025. doi: 10.1038/s41591-025-03953-8. URL https://doi.org/10.1038/s41591-025-03953-8.

[21] Zhilong Weng, Alexander Seper, Alexey Pryalukhin, Fabian Mairinger, Claudia Wickenhauser, Marcus Bauer, et al. GrandQC: A comprehensive solution to quality control problem in digital pathology. *Nature Communications*, 15(1):10685, 2024. doi: 10.1038/s41467-024-54769-y. URL https://doi.org/10.1038/s41467-024-54769-y.

[22] Mahmood Lab. UNI2-h model card. Hugging Face model repository. URL https://huggingface.co/MahmoodLab/UNI2-h. Accessed 4 November 2026; the historical downloaded weight revision remains to be verified.

[23] Tong Ding, Sophia J. Wagner, Andrew H. Song, Richard J. Chen, Ming Y. Lu, Andrew Zhang, et al. A multimodal whole-slide foundation model for pathology. *Nature Medicine*, 31(11):3749–3761, 2025. doi: 10.1038/s41591-025-03982-3. URL https://doi.org/10.1038/s41591-025-03982-3.

[24] Bin Li, Yin Li, and Kevin W. Eliceiri. Dual-stream Multiple Instance Learning Network for Whole Slide Image Classification with Self-supervised Contrastive Learning. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14313–14323, 2021. doi: 10.1109/CVPR46437.2021.01409. URL https://doi.org/10.1109/CVPR46437.2021.01409.

[25] Hongrun Zhang, Yanda Meng, Yitian Zhao, Yihong Qiao, Xiaoyun Yang, Sarah E. Coupland, et al.

DTFD-MIL: Double-Tier Feature Distillation Multiple Instance Learning for Histopathology Whole Slide Image Classification. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 18780–18790, 2022. doi: 10.1109/CVPR52688.2022.01824. URL https://doi.org/10.1109/CVPR52688.2022.01824.

[26] Zhuchen Shao, Hao Bian, Yang Chen, Yifeng Wang, Jian Zhang, Xiangyang Ji, et al. TransMIL: Transformer based Correlated Multiple Instance Learning for Whole Slide Image Classification. In *Advances in Neural Information Processing Systems*, volume 34, pages 2136–2147, 2021. doi: 10.48550/arXiv.2106.00908. URL https://doi.org/10.48550/arXiv.2106.00908.

[27] Wenhao Tang, Fengtao Zhou, Sheng Huang, Xiang Zhu, Yi Zhang, and Bo Liu. Feature Re-Embedding: Towards Foundation Model-Level Performance in Computational Pathology. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11343–11352, 2024. URL https://arxiv.org/abs/2402.17228.

[28] Takaya Saito and Marc Rehmsmeier. The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. *PLOS ONE*, 10(3):e0118432, 2015. doi: 10.1371/journal.pone.0118432. URL https://doi.org/10.1371/journal.pone.0118432.

[29] Edwin B. Wilson. Probable Inference, the Law of Succession, and Statistical Inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953. URL https://doi.org/10.1080/01621459.1927.10502953.